

UNIT 5: Networking and Open Source Software

1. Evolution of Networking

ARPANET

- First computer network
- Developed by US Department of Defense
- Foundation of the Internet

Internet

- Global network of interconnected computers

Interspace

- Future concept of interactive cyberspace communication
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2. Switching Techniques

Circuit Switching

- Dedicated path established before communication
- Used in telephone networks

Packet Switching

- Data divided into packets
 - Packets travel independently
 - Used in Internet
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3. Data Communication Terminologies

Channel

Medium through which data travels.

Bandwidth

Capacity of communication channel.

Units:

- Hz
- KHz
- MHz

Data Transfer Rate

Speed of data transmission.

Units:

- bps
 - kbps
 - Mbps
 - Gbps
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4. Transmission Media

Wired Media

- Twisted Pair Cable
- Coaxial Cable
- Optical Fiber

Wireless Media

- Infrared
 - Radio Link
 - Microwave Link
 - Satellite Link
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5. Network Devices

- Modem
 - RJ45 Connector
 - Ethernet Card
 - Router
 - Switch
 - Gateway
 - Wi-Fi Card
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6. Network Topologies

- Bus
 - Star
 - Tree
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7. Types of Network

- LAN (Local Area Network)
- MAN (Metropolitan Area Network)

- WAN (Wide Area Network)
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8. Network Protocols

- TCP/IP
 - FTP (File Transfer Protocol)
 - PPP
 - Telnet (Remote Login)
 - GSM
 - CDMA
 - GPRS
 - WLL
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9. Mobile Telecommunication Technologies

- 1G
 - 2G
 - 3G
 - 4G
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10. Email and Communication Protocols

- SMTP (Sending mail)
- POP3 (Receiving mail)
- VOIP (Voice over Internet Protocol)
- Chat and Video Conferencing Protocols

Wireless Protocols

- Wi-Fi
 - WiMax
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11. Network Security Concepts

Threats

- Virus
- Worm
- Trojan Horse
- Spam

Protection

- Firewall
- Antivirus

Legal Aspects

- IT Act (India)
 - Cyber Law
 - Cyber Crimes
 - IPR Issues
 - Hacking
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12. Introduction to Web Services

WWW (World Wide Web)

Collection of web pages connected through Internet.

HTML (Hyper Text Markup Language)

Used to create web pages.

XML (eXtensible Markup Language)

Used to store and transport data.

HTTP (Hyper Text Transfer Protocol)

Protocol used to transfer web pages.

Domain Name

Human-readable website address.

URL (Uniform Resource Locator)

Complete web address.

Website

Collection of related web pages.

Web Browser

Software used to access websites.

Web Server

Computer that stores and serves websites.

Web Hosting

Service that makes website available on Internet.

13. Web Scripting

Client Side Scripting

- VB Script
- JavaScript
- PHP

Server Side Scripting

- ASP
 - JSP
 - PHP
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14. Web 2.0

- Interactive web
- Social networking
- User-generated content

Examples:

- Social Media Platforms
 - Blogging websites
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Very Important Exam Questions

- Difference between Circuit Switching and Packet Switching
 - Define Bandwidth and Data Transfer Rate
 - Types of Transmission Media
 - Explain LAN, MAN, WAN
 - Network security threats and prevention
 - Write short note on TCP/IP and FTP
 - Explain WWW and HTTP
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Exam Tips

- Draw diagram of Star and Bus topology
- Write full forms of protocols
- Mention at least two points in difference questions
- Learn definitions clearly for 1–2 mark questions